

1 Homework

1. Suppose $(X, \mathcal{T})$ is a topological space. Let $f : X \rightarrow \mathbb{R}$ be continuous on $K \subseteq X$ compact. Then $\exists x_1, x_2 \in X$ such that $\forall x \in X$.

$$f(x_1) \leq f(x) \leq f(x_2)$$

2. Show that if a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x) = 0$ for all $x \in \mathbb{Q}$ then $f(x) = 0$ for all $x \in \mathbb{R}$.
3. Give an example of a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ and an open set U such that $f(U)$ is not open.

4. Suppose the functions $f_n(x)$ on $\mathbb{R}$ satisfy:

- (a) $0 \leq f_n(x) \leq 1$ for all $x \in \mathbb{R}$ and $n \geq 1$.
 (b) $f_n(x)$ is increasing in x for every $n \geq 1$.
 (c)

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)$$

for each $x \in \mathbb{R}$, where f is continuous on $\mathbb{R}$.

- (d)

$$\lim_{x \rightarrow -\infty} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} f(x) = 1.$$

Show that $f_n(x) \rightarrow f(x)$ uniformly on $\mathbb{R}$.

5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$. This type of function is called Lipschitz. Suppose that f is differentiable.

- (a) Show that f is continuous.
 (b) Show that

$$|f'(x)| \leq M$$

for every $x \in \mathbb{R}$.

- (c) Conversely, suppose that

$$|f'(x)| \leq M$$

for all $x \in \mathbb{R}$. Show that

$$|f(x) - f(y)| \leq M|x - y|$$

for all $x, y \in \mathbb{R}$.

6. Prove $|\cos x - \sin y| \leq |x - y|$
7. Suppose f is k -differentiable on an open interval I and $f^{(k)}(x) = 0$ for all $x \in I$. What form does f have? Prove your claim.
8. Let $f(x) = x$ for irrational x and $f(x) = 0$ rational x
- (a) Calculate the upper and lower Darboux integrals for f on the interval $[0, b]$.
 (b) Give an example of a function that is not integrable.
9. Show that if f is integrable on $[a, b]$, then f is integrable on every interval $[c, d] \subseteq [a, b]$.

Discussion

Riemann Integration

1. Let f be a bounded function on $[a, b]$. Prove that f is integrable if and only if for each $\epsilon > 0$ there exists a δ such that for all partitions P of $[a, b]$ with $\text{mesh}(P) \leq \delta$

$$U(f, P) - L(f, P) < \epsilon. \quad (1)$$

2. Prove that a bounded function f on $[a, b]$ is Darboux integrable $\iff f$ is Riemann integrable.

Cantor set

1. Let

$$A_0 = [0, 1], \quad A_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1],$$

and if we have A_n defined as a union of closed intervals, then define A_{n+1} as the union of the intervals obtained from the intervals that make up A_n by removing their middle third. For example,

$$A_2 = ([0, \frac{1}{3}] - (\frac{1}{9}, \frac{2}{9})) \cup ([\frac{2}{3}, 1] - (\frac{7}{9}, \frac{8}{9})) = [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1].$$

Set

$$C = \bigcap_{n=0}^{\infty} A_n.$$

We call C the Cantor set. Show

- (a) C is compact.
- (b) C has infinitely many points. (Hint: consider the end points of the intervals.)
- (c) $\text{int}(C) = \emptyset$.
- (d) C is totally disconnected, that is, given any $x, y \in C$ there is a disconnection U and V of C such that $x \in U$ and $y \in V$.
- (e) Every point of C is a cluster point of C (sets with this property are called perfect).